

Review retrospective:
*perturbed harmonic-Gibbs moments $\langle x^3 \rangle_h, \langle x^4 \rangle_h$ — a clean
 Gaussian-shift pass, and an import swap the build vetoed*

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Abstract

A soundness review of `Laplace/OneD/HarmonicPerturbedMoments.lean`: the cubic and quartic moments of the perturbed harmonic Gibbs measure, computed through a transport lemma that identifies the normalised perturbed law as $N(-h/\lambda, 1/(\lambda t))$. Result: **a clean fidelity pass with four landed simplifications** (a dead open and three dead locals), and one instructive near-miss — a reviewer-endorsed import swap that the build flatly rejected.

1 Setting

The fourth review in the harmonic 1D stack swept by this loop, after the regularity witness, the observable monomials, and the affine κ_3 . The file extends the I4 layer’s $k = 1, 2$ perturbed-moment closed forms to $k = 3, 4$:

$$\langle x^3 \rangle_h = -\frac{3h}{\lambda^2 t} - \frac{h^3}{\lambda^3}, \quad \langle x^4 \rangle_h = \frac{3}{\lambda^2 t^2} + \frac{6h^2}{\lambda^3 t} + \frac{h^4}{\lambda^4}.$$

2 What was reviewed

The two public theorems `gibbsExp_h_cube_harmonic_eq` and `gibbsExp_h_quartic_harmonic_eq`; the public transport lemma `gibbsExp_harmonic_h_eq_unperturbed_shift` ($\langle f(x) \rangle_h = \langle f(y - h/\lambda) \rangle_0$); and the private scaffolding (the square-completion shift `integral_shifted`, the Gaussian integrability witnesses, and the unperturbed integral closed forms `int_pow_k` for $k = 0, \dots, 4$). The informal ground truth was the file’s docstrings together with the Gaussian-moment formulas $E[X^3] = \mu^3 + 3\mu\sigma^2$, $E[X^4] = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$ at $\mu = -h/\lambda$, $\sigma^2 = 1/(\lambda t)$.

3 Fidelity / soundness findings

Sound; no defect. The mathematics is the Gaussian-shift identity: completing the square, $\frac{\lambda}{2}x^2 + hx = \frac{\lambda}{2}(x + h/\lambda)^2 - \frac{h^2}{2\lambda}$, so the perturbed weight is the unperturbed weight translated by $-h/\lambda$ times the constant $e^{th^2/(2\lambda)}$, which cancels in the normalised ratio. Every moment is then a raw Gaussian moment of $N(-h/\lambda, 1/(\lambda t))$. The reviewer independently checked each high-risk point — the square-completion *direction* (so the shifted observable is $f(y - h/\lambda)$, not $f(y + h/\lambda)$); the sign $-h/\lambda$ in the transport lemma; the five unperturbed integrals ($Z = \sqrt{2\pi/(\lambda t)}$, $\int y^2 = \sigma^2 Z$, $\int y^4 = 3\sigma^4 Z$, odd = 0), including the $(\lambda t)^{-(k+1/2)}$ exponent algebra and the $(2k - 1)!!$ double

factorials; and both headline constants against the Gaussian moments. “No fidelity mismatches found.”

4 Simplifications applied

A dead open and three dead locals. `open Real` carried nothing (every `Real`-name is qualified; bare `exp/√/π` live only in comments); `open scoped Nat` (the `!!` notation) and `open MeasureTheory` (bare `integral_*`, `volume`, `Integrable`) stay. Three locals were genuinely dead: the unused equation name `hc` on `set c := h/lam` with `hc` in `integral_shifted'`, and the local $\int e^{-t(\lambda/2)y^2} \neq 0$ in *both* moment proofs — after the denominator is rewritten to $\sqrt{2\pi/(\lambda t)}$ the closing `field_simp` consumes the $\sqrt{\neq 0}$ fact, not the $\int \neq 0$ one, so the latter’s subject is rewritten away before it could ever be fuel. Build green at 8258, no in-file warnings, public signatures byte-identical; the reviewer affirmed all four.

5 Disagreements and open questions

The build vetoed a reviewer-endorsed import swap. The `import Laplace.OneD.HarmonicCovGlobalConstancy` contributes *nothing* used directly in this file — its sole mention (`integral_with_perturbation_eq_shifted`) is in the module docstring, and the file re-derives that shift locally as `integral_shifted'`. The four harmonic symbols it actually uses all live in `Laplace.OneD.Harmonic`. So both Claude’s `grep` and the reviewer concluded the import could be swapped to the lighter `Harmonic`. *The build rejected it: Unknown identifier Threepoint.gibbsExp.* That symbol, too, arrives only transitively through `HarmonicCovGlobalConstancy` (which pulls in `Threepoint`), and `Harmonic` does not import `Threepoint`. The aggregator is transitively load-bearing for *two* dependency chains at once; a precise swap would need two imports and is a refactor beyond review scope. The swap was reverted, and the reviewer retracted on affirmation (“my earlier suggestion was too aggressive as stated”). Left as a report-only forward-tide.

6 Lessons

- **A “nothing-from-it-is-used” import can still be load-bearing for a *second* transitive chain.** The usual dead-import tell — no symbol of the imported module appears — was satisfied here, and an independent reviewer agreed; yet the import was the only path to a symbol from a *different* package (`Threepoint.gibbsExp`). Only the full build sees the whole transitive closure. This is the same “rebuild arbitrates the dead-import verdict both ways” rule, now in its sharpest form: the reviewer was confident and wrong, and so was the source `grep`.
- **A $\neq 0$ local stops being `field_simp` fuel the moment its subject is rewritten.** The two dead `hZ0_ne` facts looked load-bearing (a denominator non-vanishing hypothesis is exactly what `field_simp` wants) — but the proof first rewrites the denominator into $\sqrt{2\pi/(\lambda t)}$, after which only the $\sqrt{\neq 0}$ fact matches. The arbiter is the rebuild, not the apparent role.

7 Follow-ups

- Report-only forward-tide: refactor the `HarmonicCovGlobalConstancy` aggregator import into the two precise dependencies (`Laplace.OneD.Harmonic` + the Threepoint `gibbsExp` module).
- Remaining never-reviewed `laplace OneD` files: `QuarticBoundedPrior`, `QuarticBoundedPriorTestFn`, and the large `IntegralRemainder`.